Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/03/2011
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Manipulation of two-phase flow in low-gravity environments: drop-phase relocation by active wettability control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ithaca	**State:**	NY
Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012) NOTE: extension to 5/31/2013 is in process, per PI (Ed., 3/6/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Daniel, Susan (Cornell University) Hui, Chung Yuen (Cornell University)		
Grant/Contract No.:	NNX09AI83G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The broad technological goal of this project is to control two-phase flow in low gravity with active wettability. The scientific goal is to move droplets in a directed fashion using passive and active surface-energy gradients – the latter would yield active-wettability control. Fundamental to both tasks is the stability of the sessile drop, both natural and forced response. Accordingly, sub-tasks are identified: I: “Sessile-drop stability”; and II: “Active wettability-control”. Development of a means to electronically control wettability of a substrate surface in real time opens an opportunity to manipulate two-phase flow in the absence of earth's gravity. Condensers and separators are a crucial part of life support and power systems. It is proposed to develop a proof-of-concept device for collecting drops on a surface in low-gravity. A no-gravity cyclone separator would serve as a practical proof-of-concept of the collection technology. Motivation comes from an earth-based observation that passive chemical gradients can propel condensing stream droplets to a collector with centimeter per second speeds. Note that thermal effects are thought not to be central to the mechanism. What is needed for propelling drops is a bias, provided by the chemical gradient, and fluctuations, sufficient to overcome the contact-line tendency to pin. It is proposed to develop a tunable gradient to serve as a controllable bias and to develop a local vibration source to serve as the fluctuations. The goal is the capability to move drops on-demand in a number of directions. A scientific basis requires understanding the nature of the contact-line, contact-angle and spreading behavior, all of which can be influenced by the global drop dynamics. Computations are therefore also proposed to predict which drop oscillations will be most effective in de-pinning the contact-line and which vibrations will be most effective. Detachment will also be examined both experimentally and computationally since understanding how droplets get entrained in the bulk flow depends on the liquid/solid surface energy which will be tunable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The location of the liquid- and gas-phases in two-phase liquid/gas flows on Earth depends crucially on the strength of Earth's gravity. Many devices that perform well on Earth are unreliable for lunar and Mars missions. They falter in low-gravity. For example, the Total Organic Carbon [1] analyzer that works well on Earth is ‘currently not in use due to difficulty in bubble removal.’ Hence, monitoring the quality of on-board water supplies becomes a high risk (red-light) issue for lunar and Mars missions (risk 38 of the Bioastronautic Critical Path Roadmap (BCPR) [1]). This is not an isolated instance. Condensers are crucial to Rankine-cycle power plants [2], among other applications, and for regenerative Life Support systems (CO2 condenser/separators) [3]. Additionally, success of this research effort will likely lead to better earth-based two-phase flow systems, including condensers and other heat-exchange devices. 1. Center, J. S., Bioastronautic Critical Path Roadmap, Supporting Documents. 2. Lahey, R. T.; Dhir, V., Research in Support of the Use of Rankine Cycle Energy Conversion Systems for Space Power and Propulsion. NASA/CR-2004-213142 2004. 3. [NAP2000], Microgravity Research in Support of Technologies for the Human Exploration and Development of Space and Planetary Bodies. National Academy Press: 2000.
Task Progress:	I. Sessile-drop stability. A droplet placed on a planar substrate is held in a spherical-cap shape by surface tension. The contact-angle alpha is specified by the liquid/gas/solid triplet of materials. This is the “sessile-drop” base-state whose stability we probe. Appropriately, gravity is neglected (the “sessile” terminology, notwithstanding). Disturbances in shape from the spherical-cap move the contact-line and generate inviscid motions. Contact-line motions are governed by the “Hocking condition,” which relates deviations from equilibrium contact-angle to contact-line speed using a spreading parameter Lambda. This is an analog of the Rayleigh capillary drop problem [1] -- which has proved so useful to micro-gravity science [2]. We have found the two-parameter family (alpha, Lambda) of exact solutions to this problem [3, 4]. A sample mode-shape is shown from a perspective view and from a top view. Key predictions are being tested in experiment [5]. II. Active wettability-control. A schematic of the active surface shows how wettability will be controlled. This strategy requires a large array of reconfigurable surfaces that can switch the surface from hydrophilic to hydrophobic and back again in order to modify local surface properties. To this end, the mechanical behavior of thin, circular membranes (PDMS or latex rubber) are being tested. These membranes are “inflated” to form “balloons.” PDMS (polydimethylsiloxane) balloons are limited to about 200 % while latex rubber balloons can reach well-above 400 % strains. The effort is to characterize the mechanical properties and realize active-wettability with a small array of controlled balloons. The long-term goal is an array with diameters of 100 microns by which locally adjustable wettability can be achieved.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Bostwick JB, Steen PH. "Sessile-drop oscillations fill a symmetry-breaking periodic table." Presented at the 63rd Annual Meeting of the American Physical Society Division of Fluid Dynamics, Long Beach, CA, November 21-23, 2010. Bulletin of the American Physical Society. 2010 Nov;55(16):BAPS.2010.DFD.ER.7. http://meetings.aps.org/link/BAPS.2010.DFD.ER.7 , Nov-2010
Abstracts for Journals and Proceedings	Steen PH. "Wetting, spreading and capillary adhesion: putting shape-instability to purpose." Presented at the 63rd Annual Meeting of the American Physical Society Division of Fluid Dynamics, Long Beach, CA, November 21-23, 2010. Bulletin of the American Physical Society. 2010 Nov;55(16):BAPS.2010.DFD.NT.1. http://meetings.aps.org/link/BAPS.2010.DFD.NT.1 , Nov-2010

Abstracts for Journals and Proceedings	Chang C-T, Bostwick JB, Daniel S, Steen PH. "Mechanically-Excited Sessile Drops." Presented at the 63rd Annual Meeting of the American Physical Society Division of Fluid Dynamics, Long Beach, CA, November 21-23, 2010. Bulletin of the American Physical Society. 2010 Nov;55(16):BAPS.2010.DFD.MJ.4. http://meetings.aps.org/link/BAPS.2010.DFD.MJ.4 , Nov-2010
Articles in Peer-reviewed Journals	van Lengerich HB, Vogel MJ, Steen PH. "Coarsening of capillary drops coupled by conduit networks." Physical Review E 2010 Dec;82(6):066312. http://dx.doi.org/10.1103/PhysRevE.82.066312 , Dec-2010